RISHIKA PATEL

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SUMMARY

Computer Science undergraduate with solid foundations in DSA, network protocols, and AWS, proficient in Python, C++, Docker, and the ELK Stack, with practical experience building containerized SIEM pipelines, AI threat detection models, and vulnerability platforms, combining 400+ solved algorithmic problems with proven security credentials including a Top 2% TryHackMe ranking, CTF wins, hackathons, and OpenBugBounty disclosures.

EDUCATION

Vellore Institute of Technology — **Bhopal, Madhya Pradesh**
B.Tech in Computer Science (Specialization in Cyber Security) (CGPA : 8.84/10) — ***2023 – 2027***

TECHNICAL SKILLS

Languages: C++, Python, SQL, Bash Scripting, JavaScript, HTML, CSS
Tools & Frameworks: Docker, Git, GitHub, BurpSuite, Wireshark, Nmap, Metasploit, Nessus, Splunk, Ghidra, Scapy, Kali Linux
Cloud & Infrastructure: AWS (EC2, S3, IAM, VPC), ELK Stack, CI/CD Fundamentals, Containerization
Core Concepts & Networking: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOPs), Operating System (OS), TCP/IP, Socket Programming, Network Protocols, Secure Coding, System Optimization
Security & Ethical Hacking: Computer Networking, Vulnerability Assessment & Penetration Testing (VAPT), Reverse Engineering, Malware Analysis, Threat Detection, Web Security, OWASP Top 10

PROJECTS

**Containerized HIDS & SIEM Pipeline | ** *Python, Docker, ELK Stack, Scapy, Linux, Nmap*
- Spearheaded a containerized SIEM architecture using Docker Compose, optimizing host CPU/RAM overhead by reducing resource consumption by 40% through strategic VM migration.
- Developed a high-throughput Python HIDS sensor using Scapy to capture live traffic and stream real-time JSON alerts over TCP.

**Vulnerability Management Platform | ** *Python, SQLite, Network Security, Socket Programming*
- Engineered a modular vulnerability management platform in Python automating network reconnaissance, asynchronous port scanning, CVE signature matching, and HTTP header auditing.
- Integrated SQLite database schemas to track, correlate, and index system vulnerabilities over time, cutting manual reporting effort by 50%.
- Enhanced an automated security reporting engine that aggregates structured database findings into executive-level vulnerability assessment reports.

AI-Powered Behavioral Anomaly Detection | *Python, Scikit-learn, Scapy, Networking*
- Built an end-to-end threat detection pipeline that captures raw network packets using Scapy, converts them into behavioral features, and passes them to unsupervised models.
- Trained the system on baseline network traffic to identify subtle zero-day anomalies and multi-stage attack patterns that traditional signature-based rules miss.
- Streamed real-time JSON alert logs directly into SIEM workflows, cutting down false positives and enabling immediate incident response by 60%.

EXPERIENCE

Cyber Security Intern | Tamizhan Skills — ***Jun 2025 – Jul 2025 | Remote***
Programmed 6+ proof-of-concept Python security tools, including a custom multi-port socket scanner, file-encryption utilities and keyloggers.

CERTIFICATIONS

- Certified Cybersecurity Educator Professional (CCEP)
- Blockchain and its Applications (NPTEL, IIT Kharagpur)
- Networking Basics & Cyber Threat Management (CISCO)
- The Bits and Bytes of Computer Networking (Coursera)
- AWS Cloud Practitioner & AWS Skill Builder
- Cybersecurity Job Simulation (Mastercard & Deloitte)

ACHIEVEMENTS

- Solved 400+ LeetCode problems and actively competed in Codeforces contests with a max of 902 contest rating.
- Selected for the DSCI Cyber For Her Hackathon, qualified for Round 2 of Scripted By{Her} 2.0, Meesho Hackathon as well as in Honeywell and got an offer letter from APCSIP-2026 for a Cyber Cell internship.
- Identified and responsibly disclosed a Cross-Site Scripting (XSS) vulnerability via OpenBugBounty, earning the Fastest Fix Badge after remediation within 24 hours.
- Ranked in the top 2% globally on TryHackMe with the longest 177-day learning streak.
- Placed 355th out of 5,947 global competitors in the TryHackMe Industrial-Intrusion CTF.
- Secured 7th rank nationally in the Shell n'Zen CTF, solving complex web security and reverse engineering challenges.